

AWS GROUND STATION GLOBAL LEO SOLUTION

High Level Design Document

HLD	v1.0	AWS ONLY	CONFIDENTIAL	SPACENET-IT
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Global LEO Satellite Operator — Ground Station & Fleet Management Platform

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AWS Well-Architected | NIST 800-53 | ITAR Compliant

1. Document Control

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2. Purpose & Scope

2.1 Purpose

This High Level Design (HLD) document defines the end-to-end architecture for a global LEO satellite ground station and fleet management platform built on Amazon Web Services (AWS). It establishes the design intent, architectural decisions, component interactions, and non-functional requirements that underpin the detailed Low Level Design (LLD) documents for each subsystem.

2.2 Scope

- AWS Ground Station managed antenna network — global contact coverage
- Hybrid ground segment — on-premises antenna sites connected via Direct Connect
- VITA 49 dataflow endpoint — UDP packet receiver and CCSDS TM frame processor
- LEO fleet management platform — EKS microservices, contact scheduling, command uplink
- Real-time telemetry pipeline — Kinesis, IoT Core, Timestream, OpenSearch
- Digital twin constellation model — AWS IoT TwinMaker
- NOC operations platform — Managed Grafana, alerting, runbook automation
- Security architecture — Zero Trust, IAM, GuardDuty, Security Hub, ITAR compliance
- Multi-account AWS Organisation — CAF-aligned landing zone

2.3 Out of Scope

- Satellite bus design, payload systems, or launch operations
 - Third-party ground station network integration (beyond AWS Ground Station)
 - Application-layer mission planning software (referenced as external system)
 - End-user NOC operator training and procedures
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3. Solution Overview

The solution delivers a cloud-native, globally distributed ground station and fleet management platform for a LEO satellite operator. AWS Ground Station provides managed antenna infrastructure across 12+ global locations, eliminating the capital cost of owned antennas for global coverage. On-premises sites at owned facilities connect via AWS Direct Connect and are managed through AWS Arc. All satellite telemetry flows through a standardised VITA 49 / CCSDS protocol stack into a multi-region AWS backbone where it is decoded, stored, and made available to NOC operators and automated systems in real time.

4. Architecture Principles

Zero Trust	No implicit trust — verify every request	IAM, PrivateLink, VPC isolation, KMS
Cloud Native	Managed services over self-managed	Ground Station, EKS, Kinesis, Timestream
Event Driven	Decouple components via events	EventBridge, Kinesis, SNS, SQS
Infrastructure as Code	All infra defined in version control	Terraform >= 1.10, CDK for Lambda
Least Privilege	Minimum permissions at every layer	IAM roles, SCPs, resource policies
Immutable Infra	Replace don't patch	EKS node groups, AMI pipelines, ECR
Observability First	Logs, metrics, traces from day 1	CloudWatch, X-Ray, OpenSearch
Compliance by Design	ITAR/NIST controls embedded in arch	GovCloud option, Config, Security Hub

5. Logical Architecture

The solution is structured into five logical tiers, each with defined responsibilities and interfaces:

1	Ground Segment	Antenna contact, RF reception, VITA 49 delivery	AWS Ground Station, Direct Connect, VPN
2	Dataflow Processing	VITA 49 decode, CCSDS TM parsing, frame routing	AWS Fargate, Lambda, Event Hubs
3	Fleet Management	Contact scheduling, TC uplink, orbital state	EKS, Step Functions, DynamoDB, API GW
4	Data & Analytics	Telemetry storage, time-series, search, archive	Timestream, S3, OpenSearch, RDS Aurora
5	NOC & Visualisation	Dashboards, alerts, digital twin, automation	Grafana, TwinMaker, SNS, EventBridge

6. AWS Account & Region Structure

6.1 Multi-Account Organisation

management-prod	Root	Organisations, billing, SCPs	us-east-1
security-prod	Platform	Security Hub, GuardDuty, CloudTrail aggregation	us-east-1
shared-services	Platform	Transit Gateway, DNS, shared AMIs	us-east-1
ground-segment-prod	Mission Workloads	AWS Ground Station, TT&C, Arc servers	ap-southeast-2
fleet-mgmt-prod	Mission Workloads	EKS, telemetry processing, IoT Core	ap-southeast-2
mission-data-prod	Mission Workloads	Timestream, DynamoDB, S3, Aurora	ap-southeast-2
noc-ops-prod	Mission Workloads	Grafana, TwinMaker, Step Functions	ap-southeast-2
dr-prod	Mission Workloads	DR warm standby	us-east-1
sandbox	Sandbox	Dev/test, engineering experimentation	us-east-1

6.2 Primary Region Strategy

ap-southeast-2 (Sydney)	Primary	All production workloads, AWS Ground Station contacts
us-east-1 (N. Virginia)	Secondary / DR	DR warm standby, management plane, GovCloud gateway
eu-west-1 (Ireland)	Optional	European ground station contacts, GDPR data residency

7. Component Architecture Summary

Managed Antenna Network	AWS Ground Station	ground-segment-prod	LLD-GS-001
Hybrid Ground Sites	Direct Connect + AWS Arc	ground-segment-prod	LLD-GS-001
VITA 49 Dataflow Endpoint	ECS Fargate container	ground-segment-prod	LLD-GS-001
CCSDS TM Decoder	Lambda + ECS	ground-segment-prod	LLD-GS-001
Contact Scheduler	EKS microservice	fleet-mgmt-prod	LLD-FM-001
Telemetry Ingestor	EKS microservice	fleet-mgmt-prod	LLD-FM-001
Command Dispatcher	EKS + Step Functions	fleet-mgmt-prod	LLD-FM-001
Orbit Propagator	EKS microservice	fleet-mgmt-prod	LLD-FM-001
Fleet State Manager	EKS + DynamoDB	fleet-mgmt-prod	LLD-FM-001
Telemetry Kinesis Stream	Kinesis Data Streams	mission-data-prod	LLD-DP-001
Time-Series Store	Amazon Timestream	mission-data-prod	LLD-DP-001

Raw Frame Archive	S3 + Object Lock	mission-data-prod	LLD-DP-001
Fleet State Store	DynamoDB Global Tables	mission-data-prod	LLD-DP-001
Log Analytics	OpenSearch Service	mission-data-prod	LLD-DP-001
Digital Twin	IoT TwinMaker	noc-ops-prod	LLD-FM-001
NOC Dashboards	Managed Grafana	noc-ops-prod	LLD-FM-001
Identity	IAM Identity Center	management-prod	LLD-SEC-001
Threat Detection	GuardDuty + Security Hub	security-prod	LLD-SEC-001
SIEM	OpenSearch Security Analytics	security-prod	LLD-SEC-001
Key Management	KMS CMK (HSM-backed)	All accounts	LLD-SEC-001

8. End-to-End Data Flow

8.1 Telemetry Downlink (TM)

1	Satellite TM transmitter	RF S/X/Ka-band	Modulated RF carrier
2	AWS Ground Station antenna	RF reception	Demodulated bitstream
3	Ground Station SDR	VITA 49 over UDP	I/Q packets to dataflow endpoint
4	ECS Dataflow Endpoint	VITA 49 decode	CCSDS TM Transfer Frames
5	Lambda CCSDS Decoder	CCSDS 132.0-B-2	Space Packets (by APID)
6	Kinesis Data Streams	KPL / PutRecords	JSON telemetry records
7a	Lambda → Timestream	AWS SDK	Time-series engineering values
7b	Lambda → S3	PutObject	Raw frame archive (WORM)
7c	IoT Core → TwinMaker	MQTT/TLS	Digital twin state update
8	OpenSearch	Kinesis Firehose	Indexed for NOC search/alerts
9	Grafana	Timestream datasource	Real-time NOC dashboards

8.2 Telecommand Uplink (TC)

1	NOC operator submits command	HTTPS API Gateway	Authenticated TC request
2	Command Dispatcher validates	Internal EKS service	Sequenced TC frame

3	Step Functions dual-approval	Human task token	Approved command
4	AWS Ground Station uplink	CCSDS 232.0-B-4	CLTU uplink to satellite
5	Satellite TC receiver acknowledges	CCSDS TM frame (VC0)	ACK/NAK in next downlink
6	DynamoDB command state update	AWS SDK	Command history record

9. Non-Functional Requirements

Availability	Fleet management platform	99.99% SLA	Multi-AZ EKS, DynamoDB Global Tables
Availability	Ground contact operations	99.5% contact success	Multi-region Ground Station reservations
Performance	TM ingestion latency	< 500ms	Kinesis + Lambda direct invoke
Performance	TC uplink scheduling	< 2s API response	EKS async + Step Functions
Scalability	Telemetry throughput	10 Gbps burst	Kinesis auto-scaling shards
Scalability	Concurrent satellites	500+	EKS HPA, DynamoDB on-demand
Security	Encryption at rest	AES-256 CMK	KMS CMK all data stores
Security	Encryption in transit	TLS 1.3 minimum	ALB, API GW, IoT Core TLS policy
Compliance	ITAR data residency	US + AU regions only	SCPs, bucket policies, Config rules
Recovery	Fleet state RTO	< 1 minute	DynamoDB Global Tables active-active
Recovery	Full NOC RTO	< 1 hour	EKS warm standby DR region
Observability	Log retention	13 months	CloudWatch + S3 Glacier

10. Security Architecture Overview

Identity	IAM Identity Center SSO, MFA, PIM-equivalent JIT	NIST 800-53 AC
Network	VPC isolation, PrivateLink, Network Firewall, WAF	NIST 800-53 SC
Data Protection	KMS CMK, S3 Object Lock, TLS 1.3	NIST 800-53 SC/SI
Threat Detection	GuardDuty, Security Hub, CloudTrail, VPC Flow Logs	NIST 800-53 SI
Incident Response	EventBridge + Lambda SOAR, SSM Runbooks	NIST 800-53 IR
Compliance	Config rules, ITAR SCPs, Macie, GovCloud option	ITAR/EAR/NIST

11. External Integration Points

AWS Ground Station API	Outbound	HTTPS REST	Contact scheduling, contact profile management
On-prem antenna sites	Inbound	Direct Connect	Hybrid ground segment connectivity
Mission Planning System	Inbound	REST API	Contact requests, orbital TLE updates
Space-Track.org (18 SDS)	Outbound	HTTPS REST	TLE orbital element downloads
PagerDuty / OpsGenie	Outbound	HTTPS webhook	NOC incident alerting
ServiceNow	Outbound	HTTPS REST	Incident ticket creation
CCSDS standards bodies	Reference	N/A	TM/TC frame format compliance

12. Low Level Design Document Map

This HLD is supported by four Low Level Design documents, each covering a distinct architectural domain:

Ground Segment & Dataflow	LLD-GS-001	VITA 49 endpoint, CCSDS decoder, ECS design, hybrid connectivity	hybrid
Fleet Management Platform	LLD-FM-001	EKS microservices, APIs, state machine, contact scheduler, TwinMaker	scheduler
Data Pipeline & Storage	LLD-DP-001	Kinesis, Timestream, S3, DynamoDB, OpenSearch, data model	data
Security & Identity	LLD-SEC-001	IAM, GuardDuty, Security Hub, ITAR controls, encryption	encryption

13. Key Risks & Mitigations

AWS Ground Station regional outage	Low	Critical	Multi-region contact reservations + on-prem backup
ITAR data residency breach	Medium	Critical	SCPs + Macie + GovCloud isolation
Satellite command injection attack	Low	Critical	Dual-operator approval + command signing + PrivateLink
Kinesis shard hot partition (1 satellite)	Medium	High	Partition key strategy: spread by APID
EKS node failure during contact window	Medium	High	PodDisruptionBudgets + multi-AZ node groups
Direct Connect outage to ground site	Medium	High	VPN failover + BGP route preference
CCSDS frame decode errors (RF noise)	High	Medium	Reed-Solomon FEC + error rate alerting

14. High Level Cost Summary

AWS Ground Station (100 contacts)	\$54,000	Varies by contact duration and region
EKS + Compute (3 regions)	\$9,800	Includes node groups, Fargate, Lambda
Data & Streaming	\$11,500	Kinesis, Timestream, DynamoDB, S3, OpenSearch
Networking	\$7,200	Direct Connect, Transit Gateway, NAT
Security	\$3,500	GuardDuty, Security Hub, Macie, KMS
NOC Platform	\$2,000	Grafana, TwinMaker, SNS, EventBridge
Management & Monitoring	\$1,500	CloudWatch, X-Ray, Config, CloudTrail
TOTAL	~\$89,500	Excluding GovCloud uplift (~15%)

** Estimates based on 100 contacts/month, 50 satellites, 1TB telemetry/month. Actual costs vary.*